Residual Corrected Flow for Human-Centric Editing Across Video Shots and 3D Assets

Wenshuo Gao, Jiangyue Zeng, and Shuai Yang, *Member, IEEE,*

***Abstract*—Human-centric editing seeks to change scenes, clothing, props, and illumination while preserving the subject's identity and performance. Conditional generation and direct editing can entangle the intended changes with source reconstruction errors, whereas strong source constraints can inhibit the requested edits. We present FlowCine, a training-free framework for video and 3D editing grounded in condition consistency: unchanged conditions should reconstruct the source, making reconstruction the reference for controlled editing. Our residual-corrected flow transfers the correction required for source reconstruction to target generation, anchoring the edited trajectory to the observed input. To reconcile preservation with editability, we decouple appearance from shared structural conditions and localize residual correction and frequency transfer to identity-bearing regions. This preserves facial details and recorded motion while allowing the requested appearance changes under text instructions and independently optional visual references. We extend this formulation to sparse 3D shape and PBR appearance latents, where spatial masks and coordinate-aware frequency transfer preserve identity in the asset's coordinates during geometry and material editing. Experiments on video and 3D editing demonstrate improvements over the compared methods in source preservation, edit adherence, and visual quality. Project Page: https://gaowenshuo.github.io/FlowCine/.**

***Index Terms*—Video editing, flow matching, 3D editing.**

I. Introduction

BRINGING a visual concept to life through physical production is costly. Securing locations, preparing costumes and props, and arranging lighting require substantial resources and coordination, with each visual design imposing its own production demands. The ability to transform these elements in footage captured on a modest budget to meet more ambitious visual and production requirements would let a single recorded performance support a broader range of visual concepts, with broad applications in film-production, virtual reality, e-commercials, and creative media. Furthermore, this capability holds significant potential for future world-to-world applications that seamlessly bridge different visual worlds. Human-centric editing offers this flexibility by changing the scene, clothing, objects, and illumination while retaining the performer's identity, details, and motion. For 3D human assets, similar control would make these visual choices adjustable across viewpoints, supporting the reuse of digital performers in virtual production.

Jointly changing setting, clothing, and objects creates a conflict between preservation and editability. Directly applying these edits can change the person's face and fine details, weakening identity consistency. Conversely, strong constraints on the source appearance can preserve unwanted clothing textures, object shapes, or background content, preventing the requested changes. This trade-off becomes more challenging in video, where both the preserved details and the edited appearance must remain consistent as the person moves. An effective editing framework must protect identity-related information while allowing the regions involved in the edit to change freely according to the request.

We present **FlowCine**, a training-free framework for controllable human-centric editing of videos and 3D assets, illustrated in Fig. 1. Users specify an edit in natural language and may add background, clothing, and object images independently. A pretrained image editor combines these conditions into a target appearance, and a pretrained video/3D generative model transfers that appearance while following the source structure. This separates the choice of visual design from the recorded performance and provides the same interface for text-driven and reference-guided edits.

Specifically, our key insight is to exploit the principle of condition consistency that every change in the output is driven by a change in the target condition (*i.e.*, illumination). It implies a perfect reconstruction when there is no change in condition. Based on this principle, we introduce a residual-corrected flow mechanism that rephrases a typical flow-based generation model to a novel training-free editing model. It ensures structural consistency by directing the flow towards perfect reconstruction when the target and source conditions are identical, so that subsequent changes in the conditions like illumination variation can be reflected in the video output.

For video editing, to decouple lighting and structural conditions and enhance the model's responsiveness to conditional guidance through region-specific designs for the head, other regions within the human foreground, and the background, we propose decoupled conditions and identity-aware regional structural control. To further preserve fine-grained details in the head region, we also introduce frequency transfer, which explicitly transfers high-frequency details from the source video to the relit output. Together, these components balance identity preservation, detail consistency, and editing flexibility within one flow-based process.

For 3D editing, FlowCine encodes a supplied asset into shape and physically based rendering (PBR) appearance latents. Multi-view semantic labels define a spatial head region, and residual correction operates directly in both latent spaces. Coordinate-aware frequency transfer uses the XYZ locations of sparse features to preserve local detail. The edited latents are

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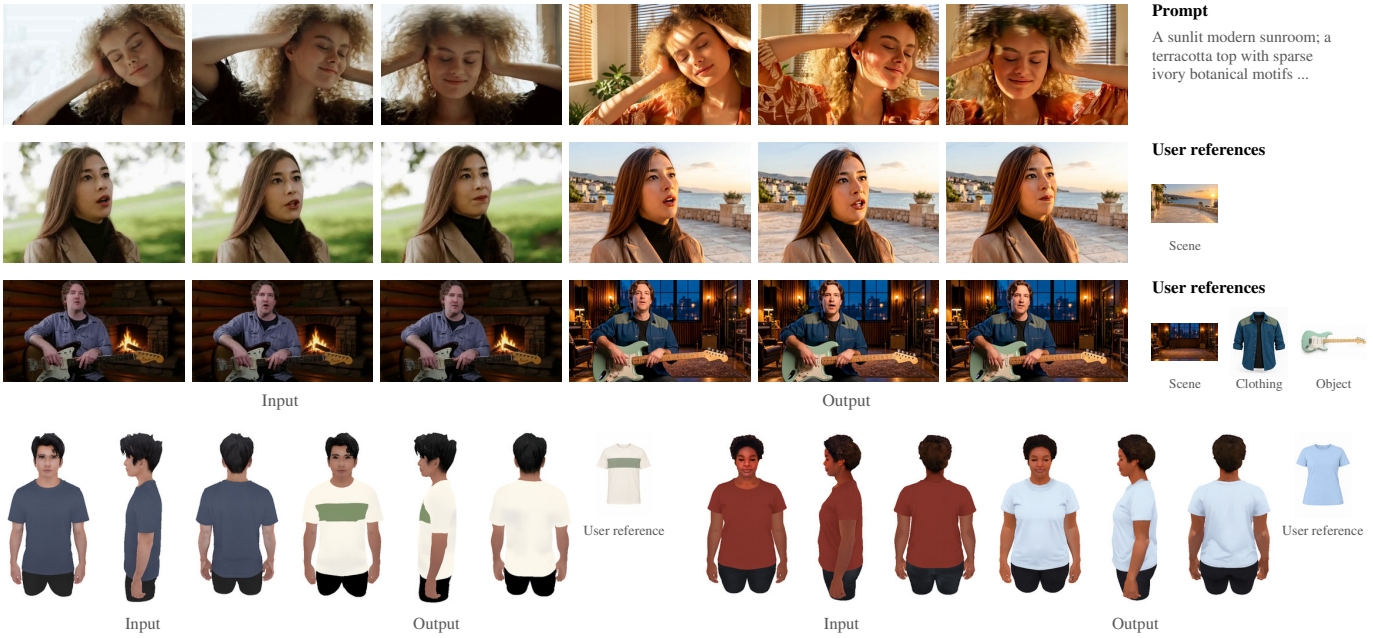


Fig. 1. Human-centric video and 3D editing. FlowCine supports text-guided and reference-guided changes to scenes, clothing, objects, and illumination while preserving human identity, details, and motion. Users can independently supply scene, clothing, and object references, and edit 3D assets with consistent appearance across viewpoints. Small thumbnails show the supplied reference images; prompt summaries accompany text-only edits.

decoded into an asset that can be viewed from new directions.

This work extends our previous work, FlowPortal [1], which studied video relighting and background replacement. FlowCine broadens the task to joint scene, clothing, prop, and illumination editing with independently optional visual references. Its technical additions include region-dependent structural control and head preservation for video, and 3D shape and PBR appearance editing with semantic spatial masks and coordinate-aware frequency transfer. We also expand the experimental evaluation with video and multi-view 3D comparisons, module ablations, and regional structure, detail, and motion metrics.

Our contributions are threefold:

- A unified human-centric editing framework covering scene, clothing, prop, and illumination changes, with independently optional visual references and a shared interface for video and 3D assets.
- A condition consistency formulation and a residual-corrected flow method that transfers source reconstruction corrections to target generation to preserve source content while enabling the requested edits.
- Region-dependent structural control and detail transfer for separating identity-related information from editable appearance, together with coordinate-aware residual correction and frequency transfer in sparse 3D latents.

The remainder of the paper is organized as follows. Section II reviews related work. Sections III and IV present residual-corrected flow for human-centric video editing and its extension to 3D assets, respectively. Section V reports comparisons and ablation studies. Multi-shot video editing is presented in the supplementary material. We discuss limitations and conclude in Sections VI and VII.

II. Related Work

A. Human-Centric Video Editing

Video editing balances requested changes against source preservation. Inversion-based approaches [2]–[5] connect editing to reconstruction, while FateZero [6], TokenFlow [7], and FRESCO [8] use attention or correspondence constraints for consistent propagation. AnyV2V [9] propagates an image-edited first frame through temporal feature injection, FastVideoEdit [10] uses consistency models for efficient sampling, and FlowEdit [11] constructs inversion-free trajectories with pretrained flow models.

VideoPainter [12] introduces plug-and-play masked-video context control, while VACE [13] unifies source videos, masks, and visual references within one conditioning interface. Kiwi-Edit [14], Lucy Edit 1.1 [15], UniVideo [16], and DITTO [17] train instruction-guided editing, while Kiwi-Edit and UniVideo also accept reference images. Omni-Video 2 [18] integrates multimodal understanding, generation, and editing. VIVA [19] uses vision-language conditioning and reward-based post-training, while EasyV2V [20] studies data construction and flexible input conditioning. Qwen-Video-Edit [21] adapt image editors to video latents, and VicEdit [22] learns from image, image-pair, and video-pair editing examples alongside text. EditVid [23] combines temporal memory, token injection, and latent blending for training-free text- and reference-guided editing. These methods face a trade-off between editing flexibility and detail preservation. Direct editing often struggles to preserve fine source details, while stronger source constraints can restrict the requested appearance changes.

B. Video Relighting and Background Replacement

Video relighting preserves surface appearance, geometry, and motion while changing illumination. The Relightables [24] enables relightable volumetric performance capture, while Neural Video Portrait Relighting [25] achieves real-time portrait relighting through temporal consistency modeling. Total Relighting [26] and SwitchLight [27] estimate normals and albedo for lighting control, while Relightful Harmonization [28] and IC-Light [29] train foreground relighting and illumination harmonization with supplied backgrounds.

RelightVid [30] and Lux Post Facto [31] train video diffusion models for consistent relighting, including portrait performances. TC-Light [32] and Lumen [33] train background replacement on long videos. IllumiCraft [34] and UniRelight [35] couple illumination with geometry and intrinsic decomposition, respectively. LumiSculpt [36] studies consistent portrait lighting, and UniLumos [37] unifies image and video relighting with physics-plausible feedback. However, collecting large-scale high-quality paired video relighting datasets is difficult and render-based datasets often lack lighting richness. As a result, they struggle to balance between the lighting conditions and the video foreground conditions. Therefore, the trained models tend to produce lighting results with insufficient richness and diversity in illumination, and lacking fidelity under complex conditions.

Training-free AnyPortal [38] and Light-A-Video [39] combine image relighting models with video models through cross-frame attention with refinement projection, and consistent light attention with progressive light fusion, respectively. However, in these two training-free methods, IC-Light is still used to relight per frame individually, introducing frame discontinuities that cannot be fully corrected by the video model. In addition, due to the lack of sufficient control in the video model, the relighting results may exhibit inconsistencies in structure and motion compared to the original video. Overcomplicated pipelines also make these methods less inference-efficient.

Our method is also training-free. By introducing the proposed residual-corrected flow and frequency transfer, we better address structural and detail consistency. To achieve stronger conditional control, we propose decoupled conditions and identity-aware regional structural control to enhance the model's ability to respond to given conditions. We also leverage residual reusing to achieve effective inference.

C. Virtual Try-On and Appearance Conditioning

Virtual try-on transfers clothing while preserving identity and pose. StableVITON [40] and IDM-VTON [41] learn clothing–body correspondence in diffusion latents and inject high-level clothing semantics and low-level visual features through separate attention paths, respectively. CatVTON [42] spatially concatenates person and clothing inputs.

For video, ClothFormer [43], CatV2TON [44], and DPIDM [45] address temporal consistency through clothing warping and temporal appearance-flow modeling, temporal concatenation for image and video try-on, and dynamic pose interaction, respectively. KeyTailor [46] and BooM-VVT [47] use selected keyframe details for clothing fidelity and motion consistency. KeyTailor improves background integrity, while BooM-VVT learns mask-free localization from image-level pseudo data and uses shared positional encoding for frame correspondence. OmniVTON [48] and PG-VTON [49] enable training-free transfer through decoupled clothing/pose constraints and patch-guided reference alignment. PROMO [50] combines flow matching, multimodal latent conditioning, and self-reference acceleration for efficient try-on.

D. Appearance Editing and 3D Priors

Appearance editing modifies object or scene appearance while preserving geometric structure and multi-view consistency. Text2Tex [51] and TexFusion [52] use image diffusion for mesh texturing through depth conditioning and shared multi-view texture denoising. Instruct-NeRF2NeRF [53] alternates image editing with radiance-field optimization, while GaussianEditor [54] uses semantic tracing and hierarchical Gaussian representations for localized scene editing.

TRELLIS [55] and TRELLIS.2 [56] provide structured latents for geometry and appearance, with rectified-flow generation, compact geometry, and material representations. AnchorFlow [57] aligns source and target flow trajectories with latent anchors for training-free shape editing, and 3D-LATTE [58] performs training-free geometry editing in 3D diffusion latents. Easy3E [59] combines voxel flow editing with normal-guided multi-view texture refinement. Hunyuan3D 2.1 [60] paints PBR materials on supplied meshes, while Steer3D [61] learns text-controlled image-to-3D editing from synthetic data. Vinedresser3D [62] and Prox-E [63] localize edits through agent-guided latent inpainting and primitive-based shape abstractions, translating instructions into spatial guidance for generative models. PriorEdit3D [64] learns feed-forward editing without paired 3D supervision by combining image-editing, semantic, and 3D priors.

However, image-based editing pipelines may introduce inconsistencies across viewpoints, while existing latent-space editing methods often lack explicit mechanisms to preserve identity-sensitive structures and fine-grained details. These limitations make it challenging to achieve flexible 3D editing while maintaining structural fidelity, appearance consistency, and identity preservation. Our method addresses these challenges in a unified training-free framework. By introducing residual-corrected flow into both shape and PBR appearance latents, we achieve consistent editing while preserving the original 3D structure and appearance details. Moreover, we leverage multi-view semantic labels for identity-aware regional control and introduce coordinate-aware Frequency Transfer to preserve fine-grained details through 3D spatial correspondence, enabling more faithful and consistent editing across novel viewpoints.

III. Residual-Corrected Flow for Human-Centric Video Editing

A. Preliminaries

FlowCine edits a source video or a textured human asset according to a text instruction and optional visual references.

Our aim is to change the requested appearance while preserving the subject's identity and source structure, motionand details. Pretrained conditional flow models supply the generative priors in the corresponding latent spaces.

a) Conditional flow models: Let z_t denote a latent at time t, where $t = 1$ corresponds to Gaussian noise and $t = 0$ to a clean representation. Under condition c, the model predicts the denoising velocity

$$V_t^c(z_t) = F_\theta(z_t, t, c), \qquad \frac{\mathrm{d}z_t}{\mathrm{d}t} = -V_t^c(z_t). \tag{1}$$

Generation follows this field from $t = 1$ to $t = 0$. In practice, we discretize the time interval into N steps using $N+1$ time points $0 = t_0 < t_1 < \cdots < t_N = 1$. Starting from $z_{t_N} = \epsilon \sim \mathcal{N}(0, I)$, an Euler step gives

$$z_{t_{i-1}} = z_{t_i} + (t_i - t_{i-1}) V_{t_i}^c(z_{t_i}), \tag{2}$$

for $i = N, \ldots, 1$. The final latent z_{t_0} is decoded into the generated output. We use this denoising convention for both video and 3D editing.

b) Sparse 3D latent representations: The 3D branch uses the sparse shape and physically based rendering (PBR) appearance representations of TRELLIS.2 [56]. A textured asset is encoded into shape features z_0^{sh} and appearance features z_0^{tex} on a common set of spatial coordinates $\mathcal{C}$. Each feature row is associated with an XYZ location in $\mathcal{C}$; the shape features encode geometry, while the appearance features encode material properties for PBR. Separate image-conditioned flow models operate on the two feature spaces. Shape is generated first, and the resulting shape features additionally condition appearance generation. The corresponding decoders convert these latent representations into a textured 3D asset that can be rendered from different viewpoints. In the following, z_t and F_θ denote the latent and flow model of the relevant modality or stage, unless specified otherwise.

B. Condition Consistency

Condition consistency expresses the relationship between a requested edit and the resulting change in the editing process. An ideal editor responds to the difference between source and target conditions while retaining attributes indentical that the instruction leaves unchanged. We formulate this requirement through two properties.

a) Directional change: When the source and target conditions differ in a particular attribute, the output should change in that attribute while preserving the remaining content. For example, when the source and target conditions differ only in lighting, the output should change accordingly while preserving structure, motion, and fine details unrelated to illumination.

b) Stability under identity: In the extreme case where the source and target conditions are identical, an ideal editing process should reconstruct the input. A zero condition difference should therefore produce a zero edit. Faithful source reconstruction provides the reference from which a controlled edit can be defined.

Direct conditional generation exposes the connection between these properties. As illustrated in Fig. 2(a), one can start from the same noise and generate z_0^{src} and z_0^{tar} under the source and target conditions. The condition describes the input incompletely and the pretrained model introduces its own reconstruction error. As a result, the source-conditioned output can differ from the input latent z_0. Consequently, editing a real recording combines the intended condition change with this source reconstruction discrepancy and even when source and target conditions are identical, the editing result fail to produce the input.

Our approach first corrects the source trajectory so that it reconstructs the given recording, then transfers the same correction to the target trajectory. Condition decoupling identifies the attributes to change, and regional preservation assigns stronger source constraints to identity-bearing content. These components connect stability under identity with directional change in human-centric editing.

C. Residual-Corrected Flow

Let z_0 be the encoded source video and ϵ a fixed Gaussian sample. We construct a restoration path from this noise to the observed input:

$$V_0 = z_0 - \epsilon, \qquad z_t = (1-t)z_0 + t\epsilon. \tag{3}$$

Following the constant velocity V_0 from $t = 1$ to $t = 0$ reconstructs z_0. This known path provides a source reconstruction reference at every noise level.

a) Consistency residual velocity: At z_t, the pretrained model predicts the source-conditioned velocity $V_t^{\mathrm{src}}(z_t) = F_\theta(z_t, t, c_{\mathrm{src}})$. Its difference from the reference velocity defines the consistency residual:

$$V_t^{\mathrm{res}}(z_t) = V_0 - V_t^{\mathrm{src}}(z_t). \tag{4}$$

Adding this residual to the source prediction yields V_0, aligning the source-conditioned trajectory with the recorded input, as shown in Fig. 2(b).

b) Transferring the correction to the target flow: We use the same residual to correct the target-conditioned velocity, yielding the residual-corrected flow:

$$V_t^{\mathrm{edit}}(z_t^{\mathrm{edit}}) = V_t^{\mathrm{tar}}(z_t^{\mathrm{edit}}) + V_t^{\mathrm{res}}(z_t). \tag{5}$$

Starting from $z_1^{\mathrm{edit}} = \epsilon$, denoising with this field produces the edited video in Fig. 2(c). The target velocity supplies the desired appearance change, while the residual anchors the process to the observed source. Under identical source and target conditions, their common initialization gives coincident trajectories and $V_t^{\mathrm{src}} + V_t^{\mathrm{res}} = V_0$, recovering the source path. When the target condition changes, the target flow introduces the requested change around this reconstruction reference.

For human-centric editing, identity and editable appearance occupy different regions. FlowCine applies the consistency residual to the head mask H:

$$V_t^{\mathrm{edit}}(z_t^{\mathrm{edit}}) = V_t^{\mathrm{tar}}(z_t^{\mathrm{edit}}) + H \odot V_t^{\mathrm{res}}(z_t). \tag{6}$$

This concentrates source preservation on facial structure and hair while allowing clothing, props, and the surrounding scene to follow the target design while it ensures that the purely generative process for new clothing textures and background structures remains free from interference.

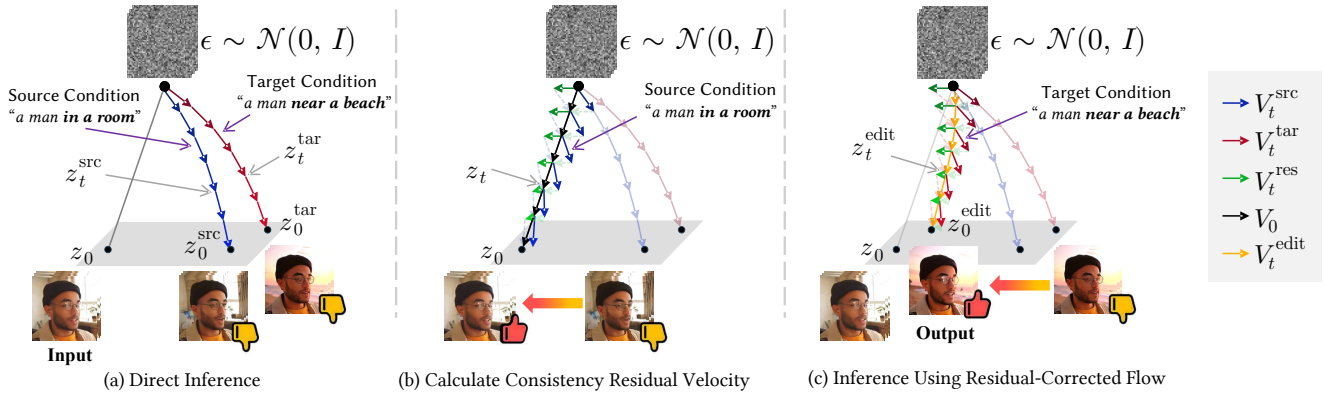


Fig. 2. **Illustration of Residual-Corrected Flow.** (a) Direct inference builds source- and target-conditioned denoising trajectories from the same noise ϵ. The generated source z_0^{src} differs from the recorded input z_0. (b) The consistency residual velocity V_t^{res} is the difference between the input reconstruction velocity V_0 and the predicted source velocity V_t^{src}, correcting the source trajectory toward the input. (c) Combining the target velocity V_t^{tar} with this residual produces the edited trajectory, preserving source structure while following the target condition. Purple arrows indicate the conditioning used to predict velocities. For simplicity, we omit the conditions other than prompt.

D. Decoupled Conditions

To guide the intended condition difference, we partition the conditioning attributes into an *edit-specific set* $\mathcal{E}$ and an *edit-agnostic set* $\mathcal{U}$ according to the instruction. The former comprises attributes to be modified, such as semantic context, background reference, clothing appearance, object appearance, and illumination, while the latter comprises attributes to be retained, including the subject's identity, structure, and motion. As illustrated in Fig. 3, source and target conditions represent the edit-specific attributes separately, while sharing the conditions that encode edit-agnostic content. This partition operates at the attribute level, that a reference image or text description can contain both editable and preserved information.

a) Separate appearance conditions: An image editor converts the instruction p_{edit} and optional visual assets $\mathcal{A}$ into a target appearance:

$$r_{\text{tar}} = G(r_{\text{src}}, p_{\text{edit}}, \mathcal{A}). \tag{7}$$

Here r_{src} is the source first frame, and $\mathcal{A}$ may be empty. The source and target images, r_{src} and r_{tar}, provide distinct appearance references for the original and requested designs. Their associated text descriptions, q_{src} and q_{tar}, specify the corresponding semantic attributes. Descriptions of preserved content remain common to both, whereas descriptions of edit-specific attributes are updated to reflect the edit. Thus, textual and visual conditions jointly express the intended change without introducing an unnecessary semantic difference in preserved content.

b) Shared structural conditions: The same source-derived structural control sequence S is supplied to both branches, providing a common geometric reference as their appearance conditions change. We combine text descriptions, image appearance, and shared source structure as

$$c_{\text{src}} = (q_{\text{src}}, r_{\text{src}}, S), \qquad c_{\text{tar}} = (q_{\text{tar}}, r_{\text{tar}}, S). \tag{8}$$

The target text and image specify the new design, while the common structural condition retains the recorded pose and performance. This decoupling aligns the condition difference with the requested appearance change and supports both text-driven and reference-guided editing through the same interface.

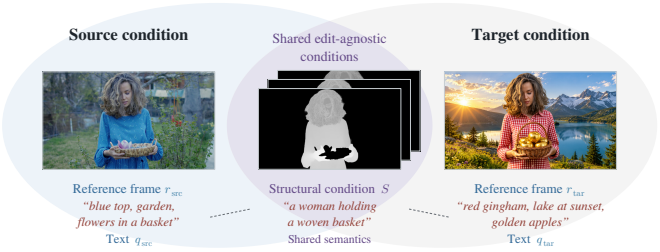


Fig. 3. Decoupled source and target conditions. Edit-specific appearance and text are specified separately for the source and target, while edit-agnostic subject semantics and the source-derived structural sequence S are shared.



Fig. 4. Source frame, person mask, head mask, and regional control. The head uses mixed depth/edge control, the remaining person uses depth, and the background control is zero.

E. Identity-Aware Regional Structural Control

Structural control follows the same division between preservation and editability. The head receives depth and edge information to retain identity-bearing geometry. The remaining person receives depth to follow the recorded pose while allowing new clothing appearance. The background control is left empty to synthesize freely. With person mask P and head mask H, the control is

$$S = H \odot S_{\text{head}} + (P - H) \odot D, \tag{9}$$

where S_{head} denotes the combination depth and edges by weighted averaging, and D denotes source depth. Figure 4 illustrates this construction.

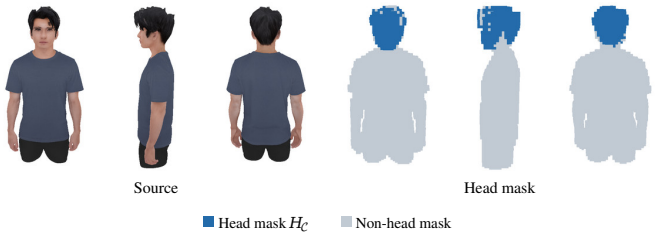


Fig. 5. Spatial head mask for 3D editing. Multiple views show the source asset and its corresponding sparse latent head mask. Blue marks the protected head mask $H_{\mathcal{C}}$; gray marks the non-head mask.

F. Fine Detail Preservation using Frequency Transfer

To further improve fine-grained detail fidelity, we introduce high-frequency transfer from the source to the edited video. Let $\mathrm{LF}(u)$ and $\mathrm{HF}(u)$ denote the low- and high-frequency components obtained by Fourier decomposition, with $u = \mathrm{LF}(u) + \mathrm{HF}(u)$. At each denoising time, the source latent z_t supplies fine detail, while the edited latent z_t^{edit} supplies the evolving appearance. We transfer source high frequencies within the head mask:

$$z_t^{\mathrm{edit}} \leftarrow (1-H)\odot z_t^{\mathrm{edit}} + H \odot \big[\,\mathrm{LF}(z_t^{\mathrm{edit}}) + \lambda\,\mathrm{HF}(z_t) + (1-\lambda)\,\mathrm{HF}(z_t^{\mathrm{edit}})\big]. \tag{10}$$

The weight $\lambda \in [0,1]$ controls the proportion of source high-frequency information. Increasing λ strengthens facial and hair detail preservation, while the target's low-frequency component carries the requested appearance. The head mask confines transfer to identity-bearing content, leaving clothing texture and the background under target control.

G. Acceleration via Residual Reuse

The consistency residual primarily supplies detail compensation that remains relatively stable across neighboring denoising steps, motivating its reuse. To accelerate video editing, we recompute the source-conditioned velocity and its consistency residual every r denoising steps, starting at the first step, and reuse the latest residual between refreshes. The target-conditioned velocity is still evaluated at every step to follow the evolving edited latent. With $r = 10$, our 50-step sampler requires 50 target and only 5 source predictions, corresponding to $1.1\times$ the flow evaluations of pure generation under the same guidance settings. Table VIII evaluates the resulting trade-off between processing time and editing quality.

IV. Residual-Corrected Flow in 3D Latents

A. Encoding a Supplied Asset

The 3D branch edits an input textured human asset through sparse shape and appearance representations. Encoding produces source features z_0^{sh} and z_0^{tex} on a common coarse coordinate set $\mathcal{C}$. The source coordinate set is retained while the feature values change, allowing the decoded asset's geometry and material appearance to follow the requested edit.

B. Appearance Conditions and a Spatial Head Mask

Source and target views provide appearance conditions c_{src} and c_{tar} for the flow models. The image editor constructs the desired appearance from a source view, the edit instruction, and an optional clothing reference. Multi-view source semantic labels are lifted onto the observed surface and associated with sparse latent coordinates, defining a spatial head mask $H_{\mathcal{C}}$. This mask assigns identity preservation consistently across viewing directions while leaving the remaining shape and appearance features editable. Figure 5 shows this spatial mask across views as an example.

C. Masked Residual-Corrected Flow

The same reconstruction principle applies to both shape and appearance latents. Let z_0 denote either source representation and ϵ shared Gaussian noise. The reference path and its denoising velocity are

$$V_0 = z_0 - \epsilon, \qquad z_t = (1-t)z_0 + t\epsilon. \tag{11}$$

The source- and target-conditioned velocities are denoted by $V_t^{\mathrm{src}}(z_t) = F_\theta(z_t, t, c_{\mathrm{src}})$ and $V_t^{\mathrm{tar}}(z_t^{\mathrm{edit}}) = F_\theta(z_t^{\mathrm{edit}}, t, c_{\mathrm{tar}})$. The consistency residual corrects the source-conditioned prediction toward this known path:

$$V_t^{\mathrm{res}}(z_t) = V_0 - V_t^{\mathrm{src}}(z_t). \tag{12}$$

Using the spatial head mask $H_{\mathcal{C}}$, the edited field becomes

$$V_t^{\mathrm{edit}}(z_t^{\mathrm{edit}}) = V_t^{\mathrm{tar}}(z_t^{\mathrm{edit}}) + H_{\mathcal{C}} \odot V_t^{\mathrm{res}}(z_t). \tag{13}$$

Spatial correction preserves identity-bearing features in the asset's 3D coordinates, while the target condition guides changes in shape and material appearance. Both stages share source-derived head mask.

D. Coordinate-Aware Frequency Transfer

Sparse feature rows have no meaningful one-dimensional frequency order. We scatter each feature channel into a three-dimensional grid at its XYZ coordinates, and filter it together with the occupancy mask O. Given a Gaussian low-pass operator L_b, we define the occupancy-normalized low-pass feature at spatial coordinate c as

$$\mathrm{LP}_{b,c}(D) = \left.\frac{L_b(OD)}{L_b(O)}\right|_c. \tag{14}$$

Normalizing by filtered occupancy preserves constants at occupied coordinates. Filtered features are gathered back at the original coordinates. Denoting the resulting sparse feature collection as $\mathrm{LP}_b(D)$, the transfer with retained high-frequency fraction η is

$$\mathcal{T}(D) = \mathrm{LP}_b(D) + \eta\,(D - \mathrm{LP}_b(D))\,. \tag{15}$$

The bandwidth b and retained fraction η determine the balance between source detail and the target appearance.

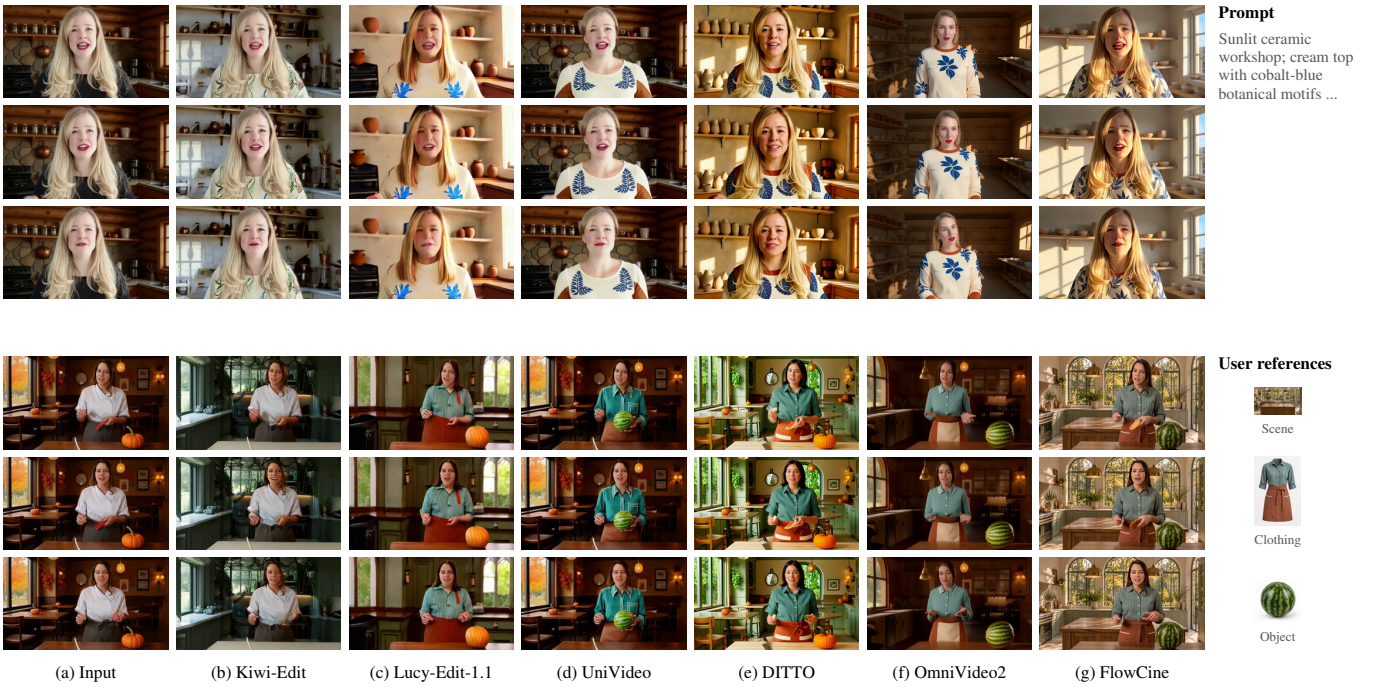


Fig. 6. Qualitative video comparisons. We compare edited sequences under text instructions and visual references, focusing on edit adherence and preservation of the original subject. FlowCine retains facial appearance and expressions while producing coherent changes to scenes, clothing, and objects.

TABLE I
QUANTITATIVE VIDEO EDITING COMPARISON.

Method	SC ↑	DC ↑	MC ↑	IC ↑	PA ↑	AQ ↑
Kiwi-Edit	0.7435	59.20	0.9883	0.327	0.298	5.148
Lucy-Edit-1.1	0.8126	61.37	0.9866	0.594	0.293	4.968
UniVideo	0.7718	59.73	0.9912	0.704	0.304	4.947
DITTO	0.6270	56.87	0.9883	0.269	0.287	5.100
OmniVideo2	0.7418	58.60	0.9532	0.333	0.303	5.188
FlowCine(Ours)	**0.8310**	**61.89**	**0.9934**	**0.719**	**0.305**	**5.325**

TABLE II
VLM-BASED VIDEO SCORING.

Method	HC ↑	IR ↑	VQ ↑
Kiwi-Edit	3.765	3.176	3.647
Lucy-Edit-1.1	3.920	2.053	3.579
UniVideo	3.864	3.591	3.545
DITTO	2.000	1.952	3.476
OmniVideo2	2.682	3.273	3.409
FlowCine(Ours)	**3.947**	**4.440**	**3.920**

E. Shape-Conditioned Appearance

The shape stage determines the edited geometry, which then conditions appearance editing. Source geometry accompanies the source appearance branch, while edited geometry accompanies the target branch. This coupling adapts the material appearance to the new shape at shared sparse coordinates. Decoding the edited representations produces a textured asset that can be rendered from new views.

V. EXPERIMENTAL RESULTS

A. Implementation Details

a) Evaluation sets: We constructed a video evaluation set containing 64 video-condition paired cases from 48 distinct video clips with 1-2 condition sets each of 49 frames at different resolutions. Meanwhile, we also conducted a 3D evaluation set contains 24 human assets.

b) Image and video backbones: The image editor used is FLUX.2-klein-9B [65]. It receives the source first frame of the video or the front view of the 3D asset, and enabled optional background, clothing, and object references, identified by role in the instruction. The video branch uses Wan2.1-Fun-V1.1-1.3B-Control [66], [67] with 50 steps.

c) Masks and structural control: FASHN Human Parser [68] initializes person and face/hair masks. Two independent MatAnyone [69] passes propagate person and head masks. Head control combines depth, Canny, and by weighted average with weights 0.8, 0.1, and 0.1. The remaining person receives depth, and background control is zero.

d) Video residual correction: Both source and target CFG scales are set to 6. Frequency transfer uses frequency threshold 0.8 with source high-frequency weight $\lambda = 0.5$ in Eq. (10). To accelerate denoising, we refresh the source prediction every 10 steps and reuse the corresponding residual for the intervening steps. The 50-step process therefore requires 50 target and only 5 source predictions, only $1.1\times$ the flow evaluations of pure generation under the same CFG, while preserving facial identity and fine details.

e) 3D implementation: TRELLIS.2 [56] runs at resolution 1,024, with 24 shape and 24 appearance steps. The source mesh is normalized to the encoder cube and voxelized with



Fig. 7. Video module ablation. Removing or altering the components leads to facial detail loss or artifacts in the edited regions. The complete model retains facial appearance and expressions while allowing coherent changes to clothing and surroundings.

TABLE III
VIDEO MODULE ABLATION STUDY.

Variant	SC ↑	DC ↑	MC ↑	IC ↑	PA ↑	AQ ↑
No residual correction	0.6509	57.1364	0.9947	0.3071	0.260	5.428
Full video correction	0.6384	**59.2576**	**0.9981**	**0.6825**	0.244	5.487
No frequency transfer	0.6119	58.6199	0.9971	0.6087	0.279	5.496
Full	**0.6580**	59.1697	0.9972	0.6705	**0.282**	**5.506**

TABLE IV
QUANTITATIVE 3D EDITING COMPARISON.

Method	SC ↑	DC ↑	IC ↑	PA ↑	AQ ↑
Hunyuan2.1	0.707	50.234	0.402	0.313	4.603
AnchorFlow + PBR	0.715	47.039	0.252	0.268	4.313
FlowCine(Ours)	**0.889**	**60.431**	**0.768**	**0.315**	**4.663**

material attributes. Shape and appearance features are matched by exact coordinate keys; a bounded nearest-cell remap handles geometry/material voxelization at source triangle boundaries. Four source views use quarter-turn azimuths, 512×768 resolution, and a 35° horizontal field of view. The frontal view supplies the image editor. Source and target images use a shared source-derived crop and independent person masks. The flow models receive image conditions. Source face/hair surface labels are mapped to latent cells through the nearest observed source surface points. Coordinate-aware filtering uses blur 0.2, high-frequency fraction 0.5. The appearance model concatenates source or edited shape features with the corresponding appearance branch. Decoded assets are exported as GLBs in the source frame with baked 2048×2048 PBR textures.

B. Evaluation Metrics

We evaluate editing quality and source preservation. Editing quality is assessed in terms of prompt alignment and aesthetics. **Prompt Alignment (PA)** measures CLIP cosine similarity between edited frames and the target prompt [6], and **Aesthetic Quality (AQ)** uses the frozen LAION aesthetic predictor V2 [70]. Source preservation is assessed through identity consistency and regional preservation of structure, detail, and motion. **Identity Consistency (IC)** measures ArcFace feature cosine similarity between corresponding source and edited faces [71]. The regional measures use common source-defined masks. Within the source head mask, **Structure Consistency (SC)** is Canny-map SSIM, while **Detail Consistency (DC)** is PSNR between LoG responses of albedo predictions [72] on the same region, using the frozen Intrinsic v2.0 model for source and output frames. **Motion Consistency (MC)** is SSIM between source and edited RAFT [73] flow components.

We also conducted measures with a vision-language model (VLM) evaluation of video editing. Given aligned source and edited frames, the edit instruction, and supplied reference images, GPT-6.1-sol assesses **Instruction and Reference Following (IR)**, **Video Quality (VQ)**, and **Human Consistency (HC)** which jointly considers structure, fine detail, motion, and identity preservation. The evaluator receives aligned source and edited frames, the edit instruction, and supplied reference images, and assigns a score from 1 to 5 for each criterion. Scores are averaged across video editing cases for each criterion.

C. Video Comparisons

We compare FlowCine with five representative video editing methods that vary in their conditioning interfaces and task coverage. Kiwi-Edit [14] supports both instruction-guided and reference-guided editing, while Lucy Edit 1.1 [15] and DITTO [17] focus on instruction-guided video editing. Uni-Video [16] and OmniVideo2 [18] integrate multimodal understanding with video generation and editing. These comparisons examine how well general video editing capabilities meet the requirements of human-centric editing, where changes to

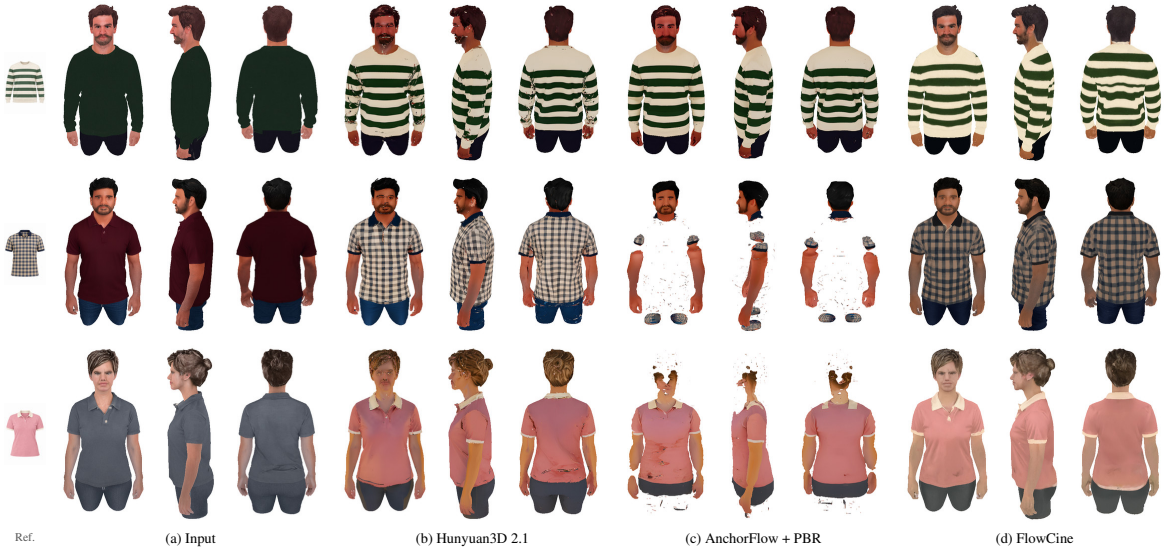


Fig. 8. Qualitative 3D comparisons. Source and edited assets are shown from multiple viewpoints to compare identity preservation and clothing-editing quality. FlowCine retains facial appearance and body structure while producing coherent clothing changes that follow the supplied references.

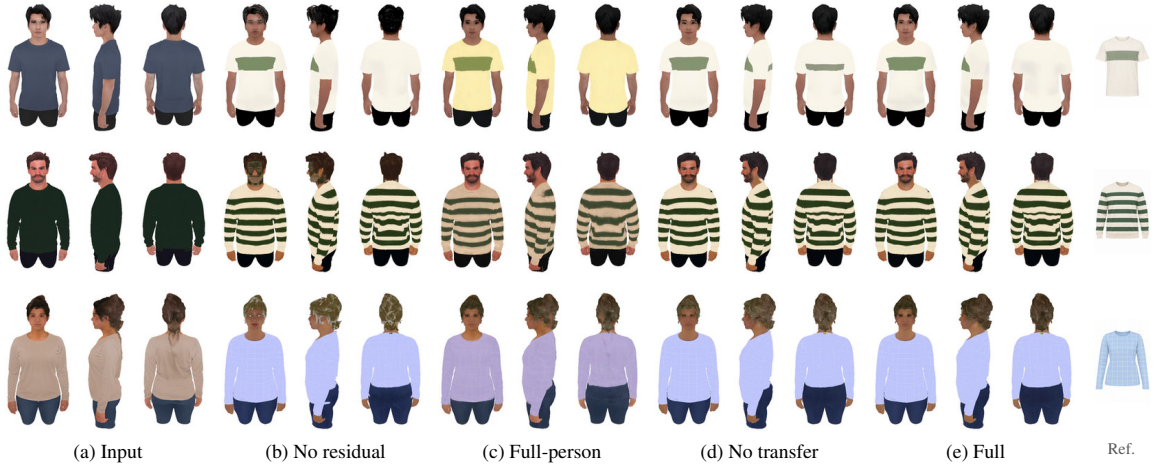


Fig. 9. 3D module ablation. The ablated variants show degradation in facial details or weaker adherence to the target clothing appearance. The full model combines identity preservation with coherent appearance changes across viewpoints.

scenes, clothing, and objects must preserve facial identity, fine details, and recorded motion.

Table I reports the comparison with other methods. FlowCine obtains the highest scores on all metrics, indicating the best overall balance between Prompt Alignment, source preservation, and visual quality. The VLM results in Table II are consistent with this comparison. FlowCine ranks the best in instruction and reference following, video quality, and human consistency. Figure 6 compares the source and method outputs, with the requested prompt and supplied user references alongside each example. The visual comparisons show that FlowCine preserves facial details while applying the requested appearance changes with the best visual quality, supporting the findings of both evaluations.

Figure 10 presents additional text-guided and reference-guided video editing results. More video results are available on our project page: https://gaowenshuo.github.io/FlowCine/.

TABLE V
3D MODULE ABLATION STUDY.

Variant	SC ↑	DC ↑	IC ↑	PA ↑	AQ ↑
No residual correction	0.804	54.451	0.276	0.299	4.579
Full-person correction	0.893	59.205	**0.759**	0.308	4.648
No frequency transfer	0.883	58.037	0.645	0.305	4.701
Full	**0.895**	**59.882**	0.758	**0.310**	**4.706**

D. Video Module Ablation

We examine three aspects of source preservation: residual correction, its spatial extent, and frequency transfer, while keeping the generation settings fixed. Removing residual correction tests the contribution of compensating for deviations from the source reconstruction path. Applying correction to the full video tests whether this compensation should also constrain regions intended for editing, including clothing and backgrounds. Removing frequency transfer while retaining

residual correction isolates the additional contribution of source high-frequency information to fine detail preservation.

The quantitative and qualitative results in Table III and Figure 7 show that removing residual correction weakens identity preservation, accompanied by changes in facial appearance and expressions. Although full video correction improves several preservation metrics, it reduces prompt alignment and introduces artifacts in edited clothing and backgrounds. This trade-off indicates that spatially unrestricted source constraints can interfere with the intended edits, supporting correction within the head region. Without frequency transfer, facial and hair details deteriorate despite the retained residual correction. These observations support the complementary roles of trajectory correction and detail transfer, with regional restriction maintaining the separation between identity preservation and appearance editing.

E. 3D Comparisons

We compare FlowCine with Hunyuan3D 2.1 [60], which provides shape generation and PBR material synthesis, and AnchorFlow [57], which performs training-free shape editing through latent anchor-aligned flows. Since AnchorFlow operates on geometry, we combine it with Hunyuan PBR painting to obtain textured 3D outputs. This setup evaluates both geometry and appearance, with particular attention to whether requested clothing changes remain consistent across views while preserving source body structure, facial identity, and surface details.

For 3D evaluation, we render the source and edited assets from four fixed viewpoints using identical cameras and unlit base-color rendering. SC is computed on Canny maps, while DC measures the PSNR between LoG responses of linearized base-color renders, both within fixed source-defined head masks. IC is evaluated only on valid frontal face pairs. Scores are averaged over valid views within each asset and then equally across assets.

Table IV shows that FlowCine achieves the highest scores across all metrics, combining strong structure, detail, and identity preservation with Prompt Alignment and Aesthetic Quality. Figure 8 compares the source and edited assets across viewpoints. FlowCine preserves facial appearance while applying the requested clothing changes consistently across views.

F. 3D Module Ablation

We further examine whether the same preservation mechanisms remain effective in sparse 3D shape and appearance representations, with generation settings fixed. Removing residual correction tests its role in retaining source geometry and appearance during editing. Extending correction from the head region to the full person tests whether broader source constraints benefit identity preservation or interfere with clothing changes. Removing coordinate-aware frequency transfer evaluates whether transferring local source details provides benefits beyond residual correction in the sparse latent space.

Table V shows that removing residual correction degrades structural, detail, and identity consistency, supporting its role in anchoring the edited asset to the source. Full-person correction provides comparable identity preservation to the full model but lower prompt alignment and aesthetic quality, suggesting that broader source constraints offer limited benefit for identity retention while restricting appearance editing. Removing frequency transfer also weakens detail and identity consistency, indicating that residual correction alone is insufficient to fully retain fine surface details. Together with the visual comparisons in Figure 9, these results support combining correction within the head region with coordinate-aware detail transfer to preserve facial appearance while accommodating the requested clothing changes.

G. Generalization across Backbones

We examine backbone generalization for video and 3D editing in Tables VI and VII. The video comparison considers Wan2.1, Wan2.2, and HunyuanVideo, while the 3D comparison considers TRELLIS.2 and Hunyuan3D 2.1. As shown, our method maintains robust performance across different architectures.

TABLE VI
VIDEO EDITING ACROSS BACKBONES.

Backbone	SC ↑	PA ↑
Wan2.1	0.8310	0.305
Wan2.2	**0.8413**	**0.334**
HunyuanVideo	0.7287	0.268

TABLE VII
3D EDITING ACROSS BACKBONES.

Backbone	SC ↑	PA ↑
TRELLIS.2	**0.889**	**0.315**
Hunyuan3D 2.1	0.572	0.275

H. Analysis of Residual Reusing

We examine the trade-off between editing quality and processing time by varying the residual reuse interval r. Table VIII summarizes Detail Consistency, Prompt Alignment, and processing time at different intervals. The results indicate that increasing the reuse step r significantly reduces inference time with minimal impact on quality.

TABLE VIII
EFFECT OF RESIDUAL REUSING.

r	DC ↑	PA ↑	Time (s) ↓
1	**62.14**	**0.307**	314
5	61.93	0.290	195
10	61.89	0.305	189
20	61.10	0.294	**170**

VI. LIMITATIONS

FlowCine is constrained by the capabilities of the underlying image editing and video generation models. As shown in Fig. 11, it struggles to produce fine details when complex backgrounds, lighting, motion, and clothing are combined.



Fig. 10. More video editing results. These examples demonstrate the flexibility of FlowCine across different subjects and various editing conditions.

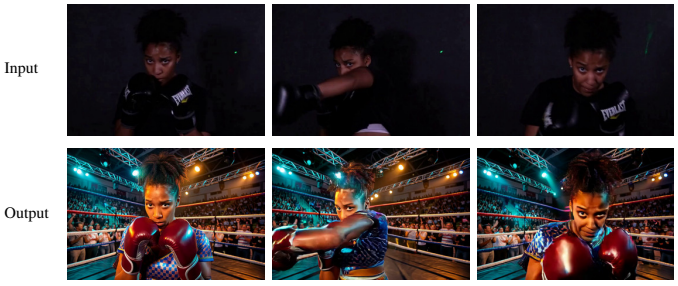


Fig. 11. Limitations on challenging video edits.

VII. Conclusion

We presented FlowCine for controllable human-centric editing of videos and 3D assets. Region-dependent structural control, residual correction, and detail transfer coordinate head preservation with scene, clothing, prop, and illumination changes. Text and independently optional visual references provide a common interface. The 3D branch edits shape and PBR latents using semantic spatial masks and coordinate-aware frequency transfer. Video and 3D comparisons examine preservation and requested appearance changes. Future work will develop more generalized editing tasks with more flexible conditions and more self-adaptive structural condition and residual correction mechanism.

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